CS316 Project: Heart disease classification

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Abstract

Heart diseases are a major global health concern, and early diagnosis can benefit both the environment and the patients. Our project used two different datasets, the first having 11 features and 70,000 records, the second containing 12 features and 299 records. We proceeded and cleaned both sets the same way by checking for duplicates, missing data, handling outliers, and checking if our data is balanced. All the models we used were classification models. We used Three classification algorithms KNN,Logistic Regression, and Decision Tree. Our work will lead to an increase in sustainability in healthcare by finding an efficient use of healthcare resources, which will lead to a decrease in energy consumption and use of materials.

1.Introduction

Our main topic that was covered in our datasets is heart disease. We used two datasets one covering cardiovascular disease and the other covering heart failure, and how earlier identification and diagnoses can help lead to longer and healthier life that will help sustainability because earlier identification and diagnoses will lead to use of less treatment and fewer patients needing extra help , This will stop the unnecessary consumption of energy, materials, and other resources associated with healthcare services. The first dataset that we used was cardiovascular disease which was a balanced data set with no missing values and no duplicates but our main issue with it was that our data was too noisy even after using IQR to remove outliers so we got an accuracy 72% from the decision tree model. So we used our second data set heart failure and got an accuracy of 81.67% from the logistic regression model.

2.Methadology

In our project, We used a Data set found by Heart Failure Clinical Records

that contains data about patients' medical records to predict likelihood of death due to heart diseases. Some of the features were age, diabetes, and blood pressure the output was whether the patient had died due to heart disease within the researchers follow up period. The Pre-processing stage includes cleaning the data and removing missing or invalid values and also handling outliers. The dataset obtained was clean with no outliers and no missing values so this stage wasn't applied. Next, the data was split into training and testing sets, this is required to ensure that the model performs the steps on unseen data and avoids memorizing patterns. We trained three classification models: logistic regression, decision tree and KNN.

1)Logistic Regression:

it uses the features to calculate probability and classifies the records according to a threshold. The process includes finding the z-score using the linear combination formula, then applying sigmoid formula to convert it into a probability between 0 and 1.

2)Decision Tree:

makes decisions by asking a series of yes/no questions about the features. Depth controls how many questions and how many splits it does. The deeper it is the more accurate but going too deep can cause memorizing patterns

3)KNN:

to classify a new record, this model calculates distance with other records and finds k nearest neighbors around it, and classifies it based on majority. A small k can be very sensitive to noise meaning incorrect data can have major effects and can lead to overfitting while a large K will consider many neighbors, maybe far away ones also and cause underfitting.

Finally, we evaluated each model using:

1-accuracy: percentage of all correct predictions of the model.

2-Percision: it answers question "when a model says 'yes', can we trust it" focusing more on predictions

3-Recall: it answers the question "off all the actual yes, how many did it find", it focuses more on actual cases.

4- F1 score which combines recall and precision.

Also, model optimization was performed on logistic regression model via CV and grid search to get the best parameters that achieve highest accuracy.

3.Summary of main insights

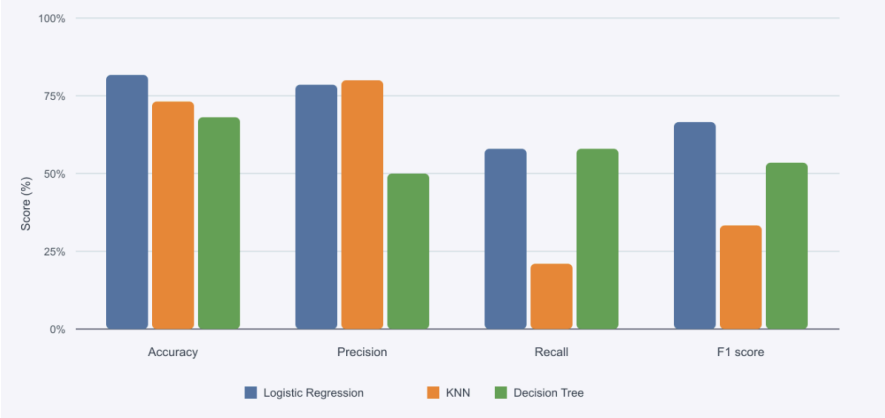
After evaluating the three classification models, several insights were observed. Logistic Regression achieved the highest accuracy of 81.67%, indicating that it correctly classified the largest percentage of patients. However accuracy alone does not indicate full performance of a model so other metrics are also evaluated. It achieved a precision of 78.57%, showing good

reliability in its positive predictions, but a low percentage in recall indicates that model misses many actual positives which can be bad for medical systems.



This graph compares the performance of logistic regression models on different metrics.

Limitations for logistic tree include poor performance on high complex situations as seen on our first data sets, it assumes a linear relationship between the features and outcome so performs badly on non linear features.



This graph compares the performance of all three models, KNN achieved the highest precision of 80% indicating it produced fewer false alarms, but had a very low recall of 21.05%, meaning it

missed many actual positive cases. KNN disadvantages include being slow with large datasets because it must calculate distance with many examples, also is affected by choice of K-value. Overall, Logistic Regression performed best in terms of accuracy

4.Conclusion

Our project helped us explore how different datasets and classification models affect accuracy and how models run. Our main issue that we faced during our project was our inability to reach an accuracy of 80% or higher when we used our first data set. Even after cleaning and removing outliers in it. Our dataset was still too noisy. That was why we changed our dataset. We were able to reach an accuracy over 80% using it. This shows the importance of choosing a proper dataset. We used three classification models KNN,Logistic Regression, and Decision Tree. Our best performing model in our second dataset was a logistic regression model that gave us an accuracy of 81.67% which is almost 10% more than our original dataset which gave us 72%. Overall, our project can help the healthcare field be more sustainable by being able to help with early diagnosis of different heart diseases which will lead to decrease in wasting material and a higher chance of survival for patients by being able to find risks early on.

5.Use of AI statement

Our main use of AI was for assistance in drawing our graphs, to compare our algorithms and how they performed. AI was given prompt without evaluation results and was asked to represent them in graph forms. AI's assistance was also used to clear up some confusions faced in the coding phase. The AI tool that we used was ChatGPT.

6.References

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